Understanding & Boost Consumer Behavior for Online Shopping

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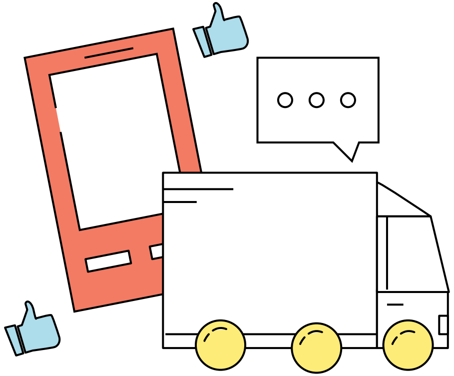
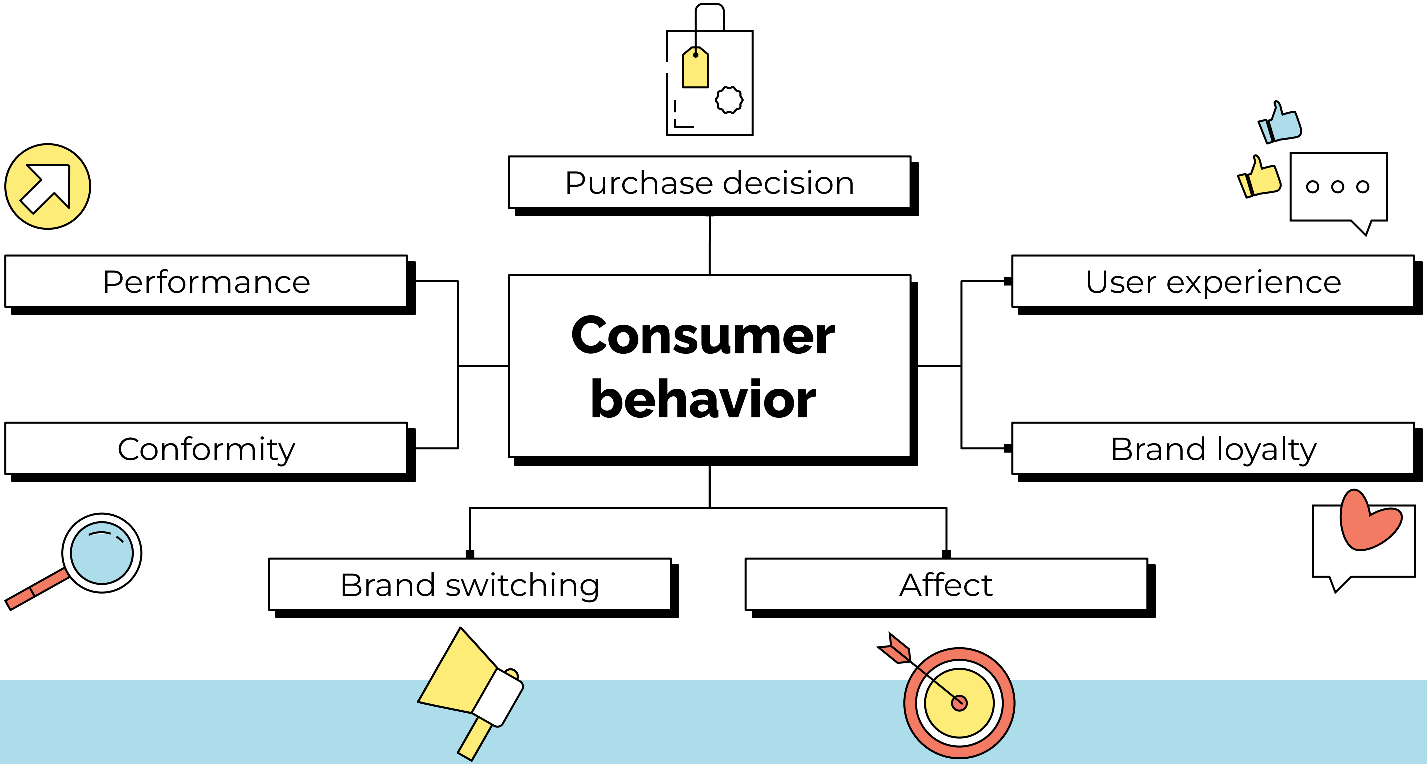


Introduction



Objectives

Understanding

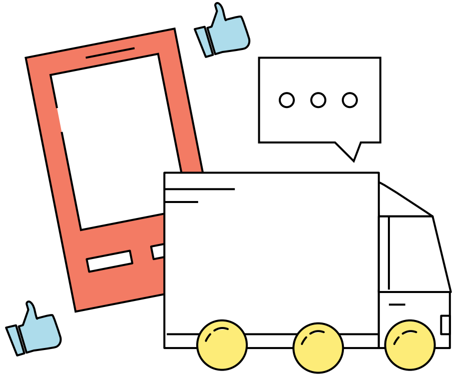
Understand customer behavior and engagement patterns by analyzing how customers browse and interact to identify what keeps them engaged.

Prediction

Use machine learning to predict and improve user engagement, improve the shopping experience and marketing effectiveness.

Clickstream Data for Online Shopping Dataset records clickstream activity from an online store selling clothing for pregnant women over five months in 2008. It includes product category, photo position, model photography, and product price.

02 Methodology



Methodology

Data Setup & EDA

Preprocessing, Exploratory Data Analysis

Data Engineering

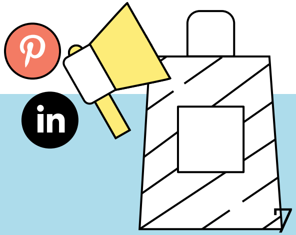
Feature Engineering, Baseline Model

Model Tuning

Cross Validation, Grid Search, Ensemble Models

Feature Impact

Feature Importance



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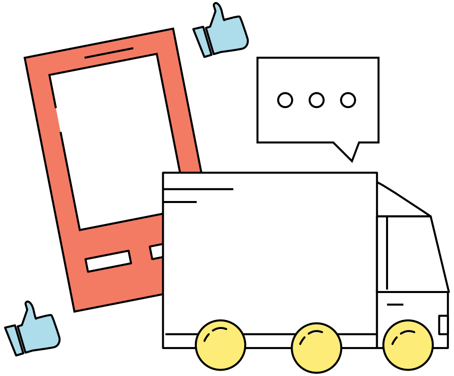
Analysis & Insights



Step 1: Data Setup & EDA

Product Price Influences
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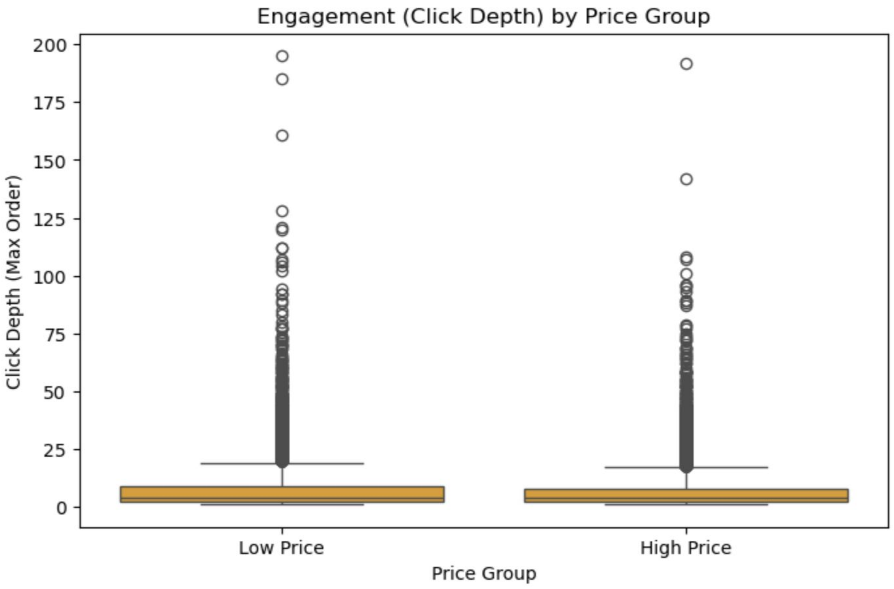
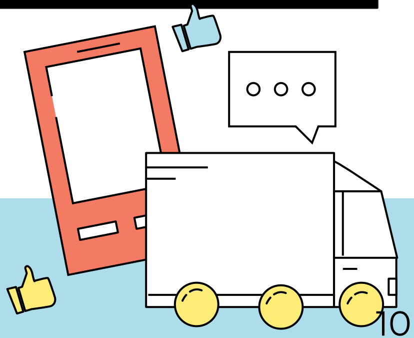
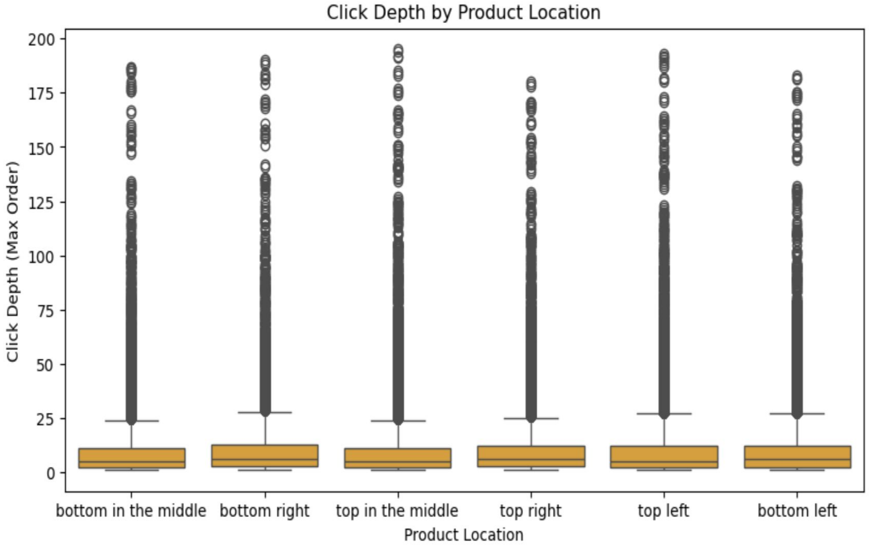


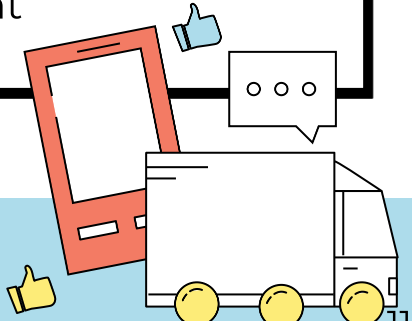
- ...s with **lower-priced products show deeper click behavior** (avg. 7.03 clicks (low price) vs 6.75 clicks (high price))
- The difference is **small but statistically significant** (t-test, $p < 0.05$)



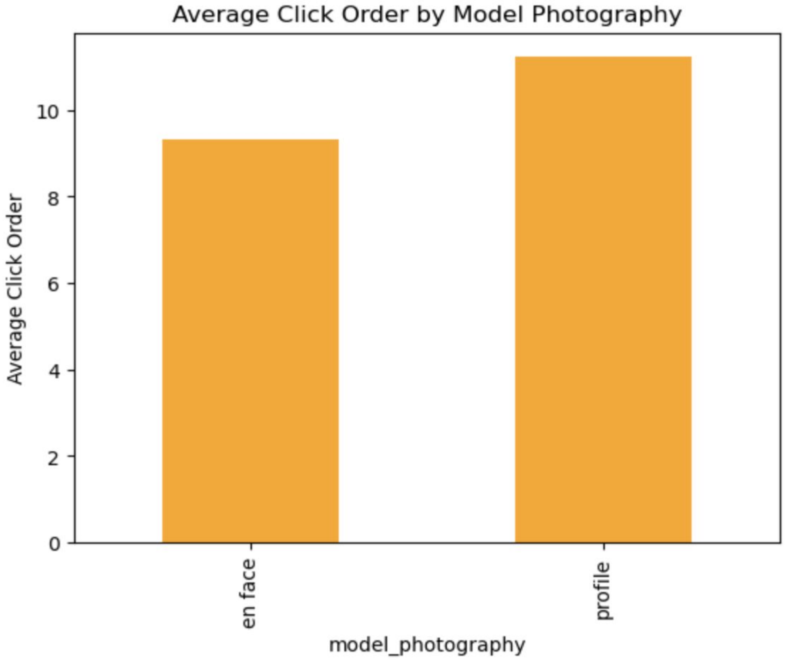
Product Presentation Affects Engagement



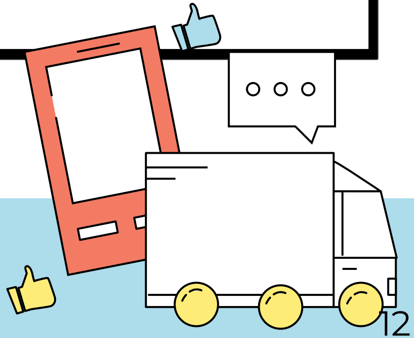
- Top middle (195) and top left (193) show the highest engagement
- Bottom positions generally have lower engagement (183–190)
- **Top positions show slightly higher engagement**, suggesting that visually prominent product placement leads to increased user engagement



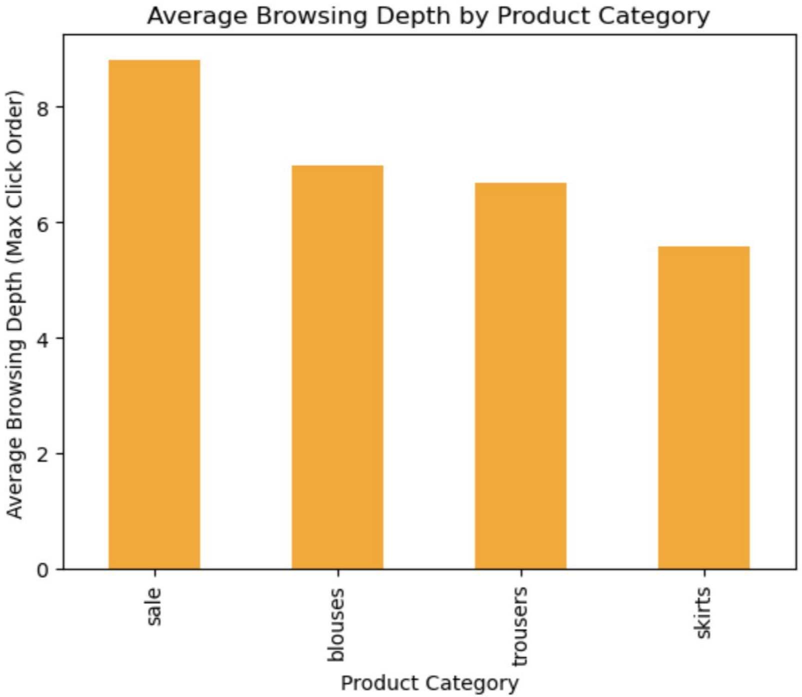
Model Photography Influences Early Clicks



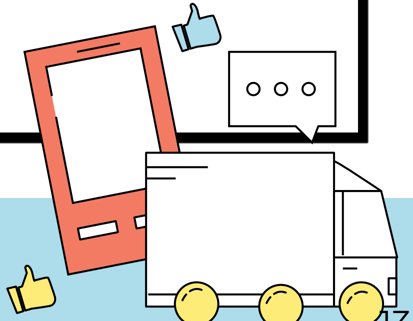
- Profile images have a higher mean click order
- En face images have a lower mean click order
- Model photography style influences initial user engagement, with **en face images attracting clicks earlier** in the session



Category Drives Browsing Depth



- Sale items drive the deepest engagement (avg. 8.81 clicks)
- Blouses & trousers show moderate interest (avg. 6.7–7.0 clicks)
- Skirts show shallow browsing (avg. 5.57 clicks)
- Browsing depth differs by category, and **sale items get the most attention**



Seasonality Affects Browsing
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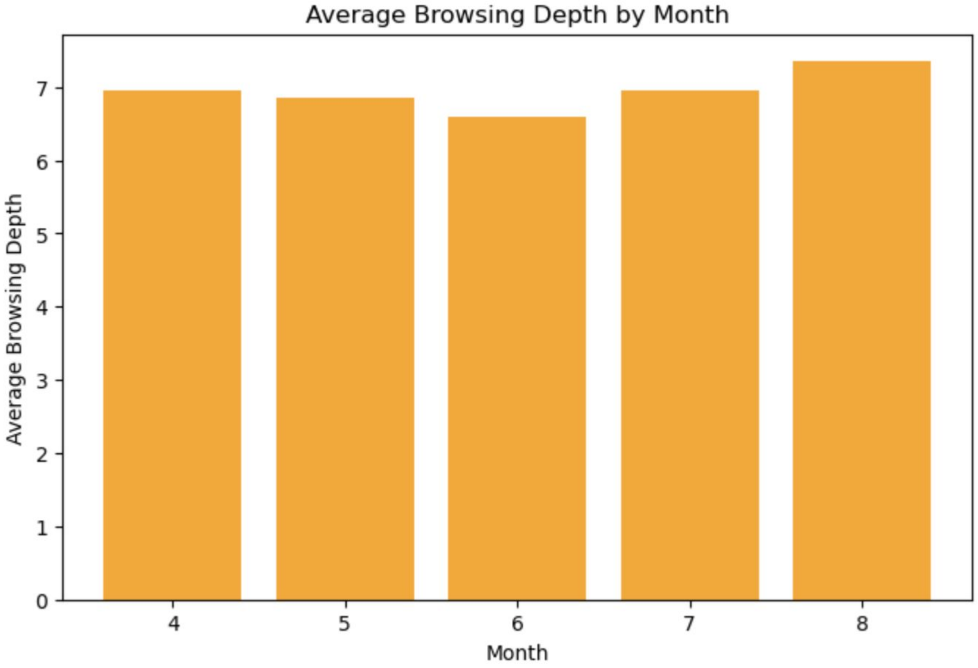
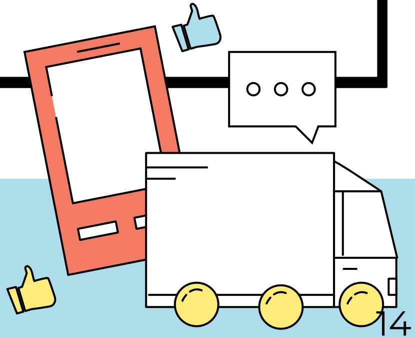
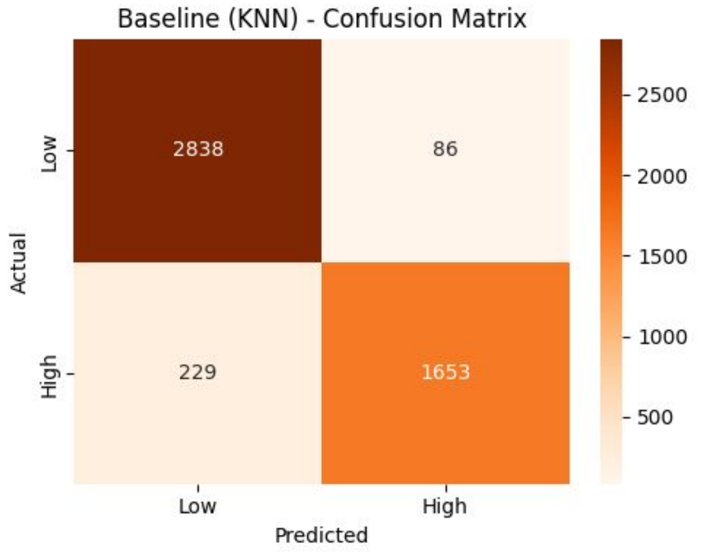


- Engagement stays stable during most of the year
- Highest browsing depth occurs in August
- **Seasonal factors affect online shopping behavior**



Baseline Model: KNN

- **Data split**: 80% training / 20% testing
- **Baseline model**: k-Nearest Neighbors (KNN)
- **Performance**: accuracy 93%
- KNN provides a **strong baseline**, especially for identifying low-engagement sessions, with room to improve recall on high engagement.

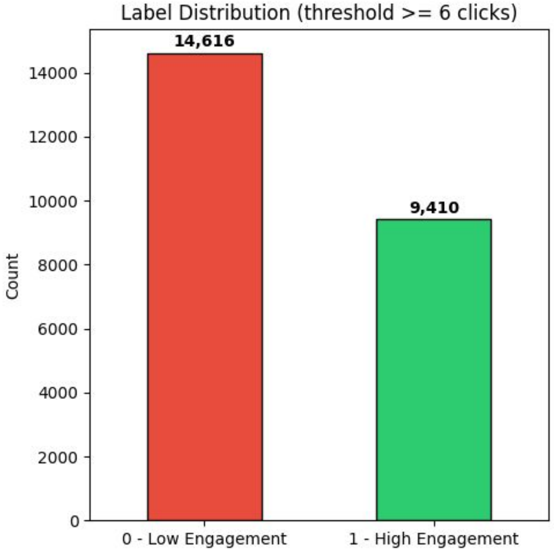


Step 2: Data Engineering

Feature Engineering

Feature Engineering

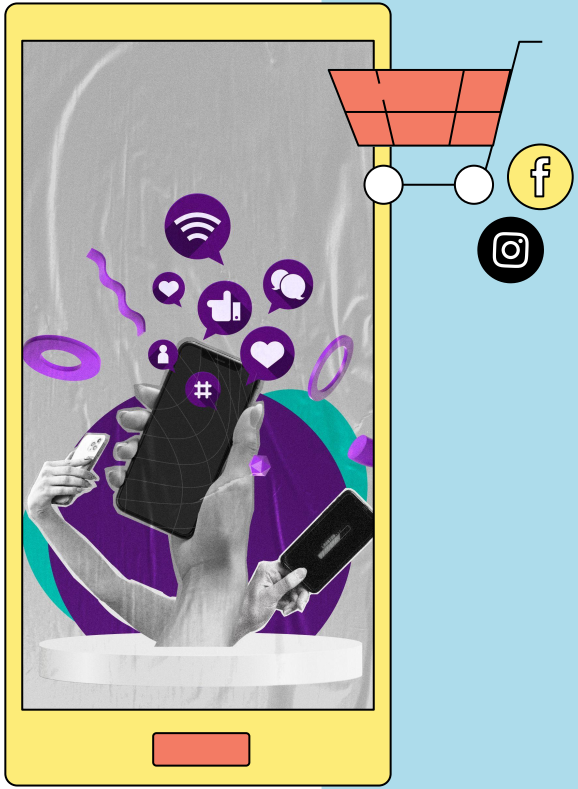
- Convert the click-based dataset into a **session-based dataset** by grouping events using the session ID
- **Create additional features** to better represent user behavior within a session (e.g., '*dominant_category*', '*dominant_colour*', '*skirt_clicks*', *etc.*)
- Transform features:
    - Categorical features using One-Hot Encoding
    - Numerical features using Standard Scaling

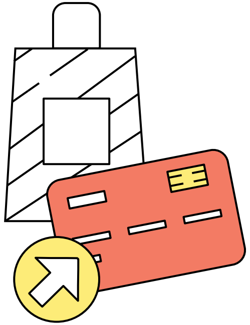


Label Creation

- Create a **binary engagement label** based on the total number of clicks per session
    - Label 0 (low engagement): sessions with few clicks
    - Label 1 (high engagement): sessions with many clicks (top 40%)
- Define engagement threshold (6 clicks) using click distribution

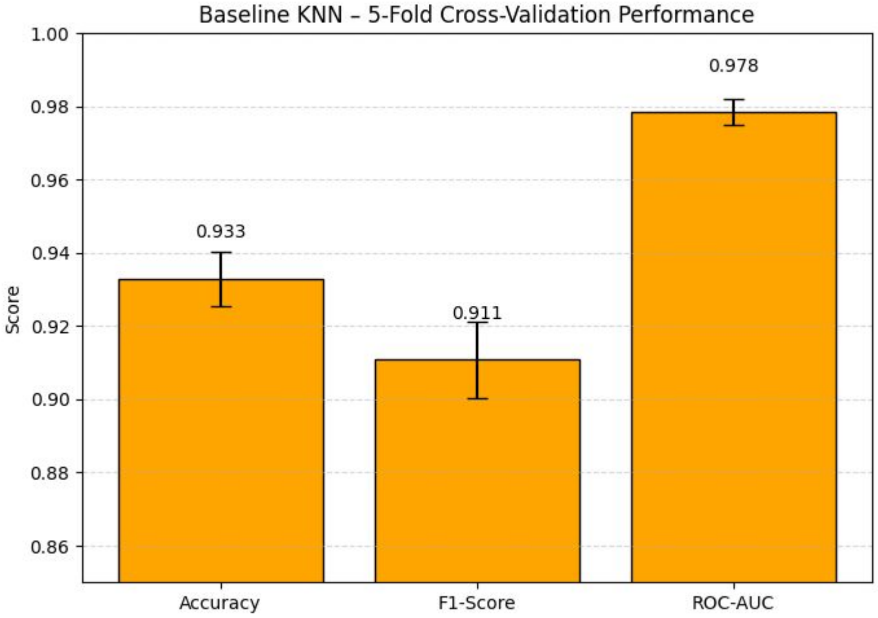
Step 3: Model Tuning



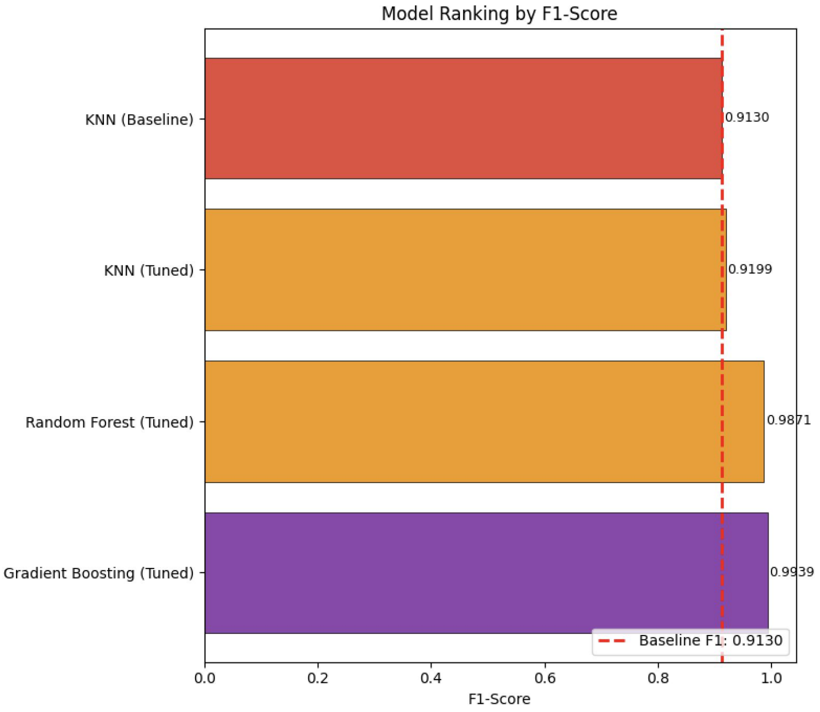


Cross-Validation

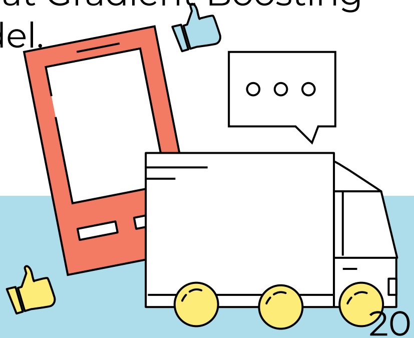
- **Method**: 5-fold cross-validation
- **Model**: KNN (baseline)
- **Results**
  - Achieves 93.3% accuracy and F1-score of 91.1%, indicating balanced performance across classes
  - The model is stable and generalizes well to unseen data
- Confirms KNN as a **reliable** and **solid baseline** for further model comparison



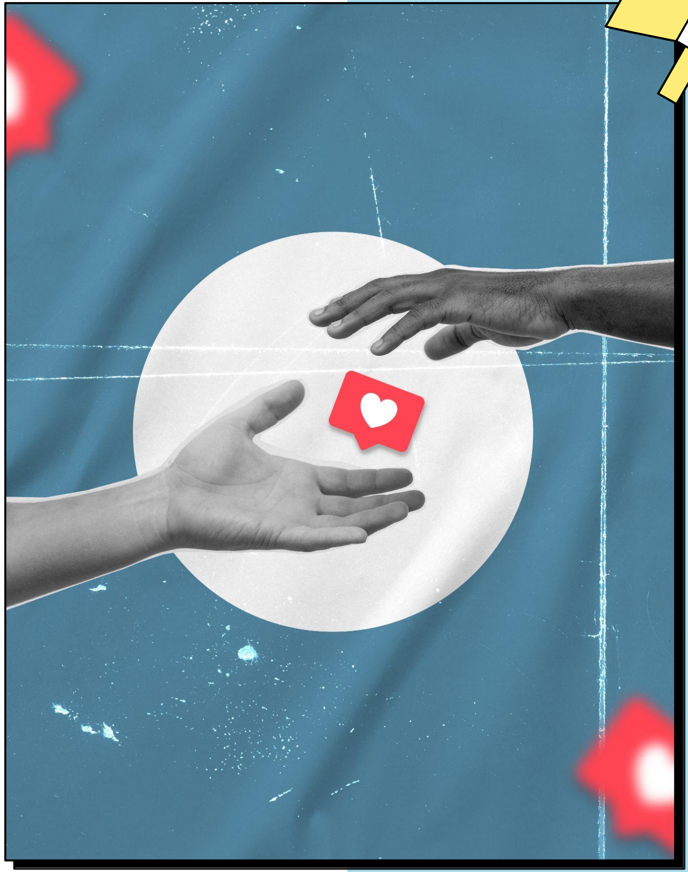
Hyperparameters Tuning



- **Method**: 5-fold Stratified Cross-Validation using GridSearchCV
- **Models tuned**: 3 classification models
- Results
  - Optimal hyperparameters identified for each model
  - **Best-performing model: Fine-tuned Gradient Boosting**
  - Achieved **99.4% accuracy**, significantly outperforming the baseline
- **Hyperparameter tuning greatly improves performance** and shows that Gradient Boosting is the best-performing model.



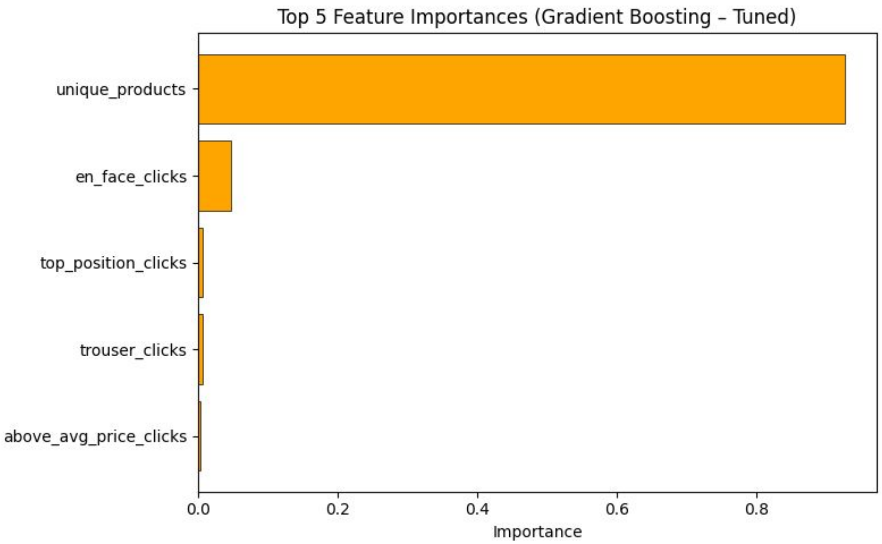
Step 4:
Feature Impact



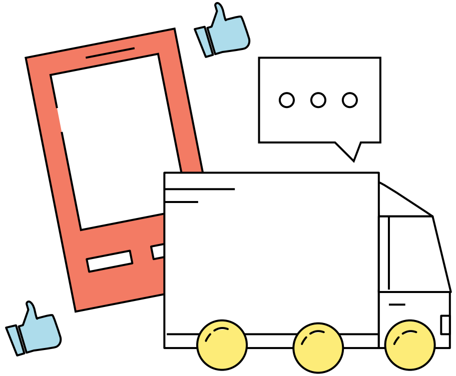
Feature Importance

Extract feature importance from the best fine-tuned Gradient Boosting model, based on how much each feature reduces prediction error across all trees

- *'unique_products'* (~93%) is the **most influential feature**, indicating that the diversity of products a user interacts with strongly drives predictions
- *'en_face_clicks'* (~4.7%), capturing **user engagement with face-category products**
- Other features (<1%), such as position- or category-specific clicks have minimal individual impact, suggesting the model relies more on overall product diversity than fine-grained click patterns



04 Conclusion



Conclusion

- **User engagement in online shopping can be predicted** effectively using machine learning
- **Price, product position, image style, category, and seasonality** influence browsing behavior
- **Product diversity viewed is the strongest factor** for high engagement
- Insights can help improve:
    - Product recommendations
    - Page layout and design
    - Marketing strategies
- **Data-driven decisions can improve the online shopping experience**

Resources

- Github Final Project
- Clickstream Data for Online Shopping Dataset
- Slide Templates

Thanks!

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